p-values and confidence intervals

A reading card. Keep it beside you when you look at the Estimate, interval and p-value in a results table.

1 Two numbers that answer two questions

P-VALUE

Is zero still a believable answer?

We start with what we want to prove wrong, for example, an assumption that the true effect is zero. A small p-value means data as extreme as ours would be unusual if the effect were zero. In other words, the data give us good reason to doubt that the effect is zero. It does not tell us that our estimate is the true effect. The confidence interval shows the range of effects that fit the data.

It does not tell you how big the effect is, whether the programme works, or whether the comparison behind the number is fair.

95% CONFIDENCE INTERVAL

Which effect sizes fit the data?

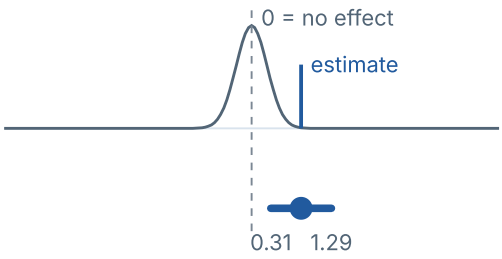
A range around the estimate that shows its uncertainty, given the model and its assumptions. A narrow range is a precise estimate. A wide range is an imprecise one.

It does not tell you where each business will land. It describes the average effect, not individual businesses.

2 How they fit together

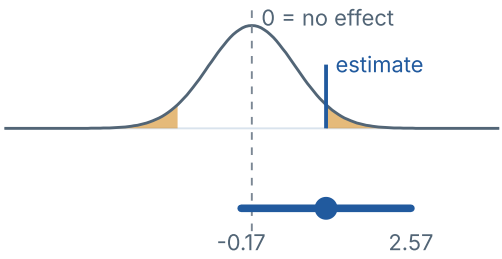
The curve shows what we would expect to see if the true effect were zero. The shaded tails are the p-value. The bar underneath is the confidence interval.

Precise estimate, interval excludes 0



Estimate **0.8** p-value **0.001** 95% CI **0.31 to 1.29**

Noisy estimate, interval includes 0



Estimate **1.2** p-value **0.086** 95% CI **-0.17 to 2.57**

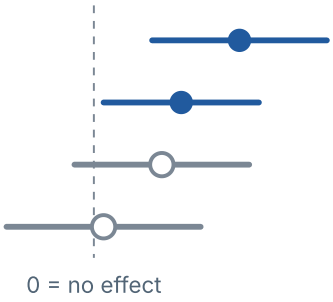
Interval excludes 0
The p-value is below 0.05 (for a 95% interval).

Interval includes 0
The p-value is above 0.05. Zero stays a believable answer.

Wide interval
The estimate is imprecise, whatever the p-value says.

3 Reading several results at once

Four hypothetical results. Filled dots have an interval that excludes 0. Hollow dots have an interval that crosses 0.

Result	Estimate and 95% interval	Estimate [interval]	p-value
Result 1		1.5 [0.6, 2.4]	0.001
Result 2		0.9 [0.1, 1.7]	0.03
Result 3		0.7 [-0.2, 1.6]	0.13
Result 4		0.1 [-0.9, 1.1]	0.85

4 Find them in the table

The GreenWaste before-and-after result for participating businesses, as in the Session 1 table.

Row	Estimate	95% interval	p-value
1 Change in annual waste cost (AED)	2 -669	3 -684 to -654	4 < 0.001

1 Name the row and its units. Negative means lower cost.

2 The estimate: its size and sign.

3 The interval: how precise the estimate is.

4 The p-value: is zero still believable?

5 Say it in a sentence

Pattern: **[Outcome] [rose or fell] by [estimate] [units] (95% CI: [low] to [high], p = [p]). [What the interval allows.] [What the design allows.]**

INTERVAL EXCLUDES ZERO (GREENWASTE)

Annual waste cost fell by 669 AED per business (95% CI: 654 to 684 AED, p < 0.001). The change is statistically significantly different from zero: the data are not consistent with no change. The comparison is the same businesses before and after, so it does not by itself show that GreenWaste caused the fall.

INTERVAL INCLUDES ZERO (HYPOTHETICAL RESULT B)

The estimated reduction was 1,100 AED (95% CI: -100 to 2,300 AED, p = 0.072). The interval includes zero, so we cannot rule out no reduction. It also includes reductions well above our 1,000 AED rule, so the evidence is inconclusive.

PRECISE BUT SMALL (HYPOTHETICAL RESULT A)

The estimated reduction was 50 AED (95% CI: 30 to 70 AED, p < 0.001). The estimate is precise, and every value in the interval is far below our 1,000 AED rule.

6 Common slips and what to say instead

Avoid	Say instead
p = 0.072, so GreenWaste has no effect.	The data do not rule out zero, and the interval also allows large reductions.
p < 0.001 means there is a 99.9% chance the programme works.	The p-value starts from zero effect. It is not the probability that the programme works.
p < 0.001, so the saving is big.	Compare the estimate and its interval with the decision rule (1,000 AED).
95% of businesses saved between 654 and 684 AED.	The interval describes the average change, not each business.

Try with AI. Paste the table row and ask: "Explain this row for a director. Say what the p-value does not tell us, and ask me which decision rule applies before you conclude." Then check each number against the table yourself.

All figures are fictional training data from the GreenWaste case. Results A, B and Results 1 to 4 are hypothetical.